

JESS Thermodynamic Database v8.9

Tantalum

24-Jan-23

Reaction No. 40400							
Ta+5_OH-1(2) + Oxalic-2 = Ta+5_OH-1(2)_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1.5	Self Medium	lgK:11.1(3SF)	2	MSL	30107
Note (1): Ta+5_OH-1(2) improbable species							
2	18:20			lgK:11.10(4SF)	2	MSL	931
3	25	0	Inf. Dilution	lgK:11.1(3SF)	1	ELC	

Reaction No. 40401							
Ta+5_OH-1(2) + 2<Oxalic-2> = Ta+5_OH-1(2)_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20			lgK:18.52(4SF)	2	MSL	931
2	25	0	Inf. Dilution	lgK:18.5(3SF)	1	ELC	

Reaction No. 40402							
Ta+5_OH-1(2)_Oxalic-2 + OH-1 = Ta+5_OH-1(3)_Oxalic-2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1.5	Self Medium	lgK:13.3(3SF)	2	MSL	30107
Note (1): Ta+5_OH-1(2) improbable species							
2	18:20			lgK:13.33(4SF)	2	MSL	931
3	25	0	Inf. Dilution	lgK:13.4(3SF)	1	ESC	

Reaction No. 43264							
Ta+5_OH-1(4) + Tartaric-2 = Ta+5_OH-1(4)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20:22	3	Unknown	lgK:0.72(0.04SD)	4	MDS	906
Note (1): Ref: G.N. Shabanova, N.A. Skovik Z. obshchei Khim 1972 42 204 but this reference does not contain these data							
2	25	0	Inf. Dilution	lgK:1.0(2SF)	1	EES	

Reaction No. 43265							
Ta+5_OH-1(4) + H+1(-1)_Tartaric-2 = Ta+5_H+1(-1)_OH-1(4)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	20:22	3	Unknown	lgK:8.83(0.08SD)	4	MDS	906
Note (1): Ref: G.N. Shabanova, N.A. Skovik, Z. obshchei Khim 1972, 42, 204 but this reference does not contain these data							
2	25	0	Inf. Dilution	lgK:9.(1SF)	1	EES	

Reaction No. 43266							
Ta+5_OH-1(4) + H+1(-2)_Tartaric-2 = Ta+5_H+1(-2)_OH-1(4)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20:22	3	Unknown	lgK:15.3(0.2SD)	4	MDS	906
Note (1): Ref: G.N. Shabanova, N.A. Skovik, Z. obshchei Khim 1972, 42, 204 but this reference does not contain these data							
2	25	0	Inf. Dilution	lgK:15.5(3SF)	1	EES	

Reaction No. 43704							
Ascorbic-2 + Ta+5 = Ta+5_Ascorbic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:9.5(2SF)	0	MSP	906
2	23	0	Unknown	lgK:9.5(2SF)	0	MSP	30601
Note (2): Ta+5 unreal species							

Reaction No. 44881							
Ta+5_OH-1(2) + EDTA-4 = Ta+5_OH-1(2)_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1.08	K2SO4	lgK:33.6(3SF)	0	MPL	906
Note (1): Error in reacting species. Ta+5_OH-1(5) in original publication							

Reaction No. 53173							
F-1 + Ta+5 = Ta+5_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:6.37(3SF)	0	MIS	5766
Note (1): Ta+5 unreal species; Ta+5_F-1 improbable species							

Reaction No. 53174							
2<F-1> + Ta+5 = Ta+5_F-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:11.85(4SF)	0	MIS	5766
Note (1): Ta+5 unreal species; Ta+5_F-2 improbable species							

Reaction No. 53175							
$3<F-1> + Ta+5 = Ta+5_F-1(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:16.03(4SF)	0	MIS	5766
Note (1): Ta+5 unreal species							

Reaction No. 53176							
$4<F-1> + Ta+5 = Ta+5_F-1(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:19.63(4SF)	0	MIS	5766
Note (1): Ta+5 unreal species							

Reaction No. 53177							
$5<F-1> + Ta+5 = Ta+5_F-1(5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:23.29(4SF)	0	MIS	5766
Note (1): Ta+5 unreal species							

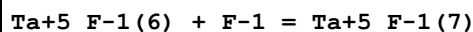
Reaction No. 53178							
$7<F-1> + Ta+5 = Ta+5_F-1(7)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:30.21(4SF)	0	MIS	5766
Note (1): Ta+5 unreal species							

Reaction No. 53727							
$Ta+5_F-1(3) + F-1 = Ta+5_F-1(4)$							
Note: Reacting species unknown, probably hydroxylated							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:5.90(3SF)	2	MQH	5398
2	25	3	HClO4	lgK:5.86(3SF)	2	MDS	931
3	25	3	HClO4	lgK:5.86(3SF)	2	MDS	30162
4	25	3	NaClO4	lgK:5.86(3SF)	2	MDS	437

Reaction No. 54409							
$Ta+5_F-1(5) + F-1 = Ta+5_F-1(6)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0.25	HNO3	lgK:3.16(3SF)	3	MSL	437

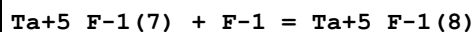
2	25	1	(H,Na)ClO ₄	lgK:3.6 (2SF)	4	MAX	140
3	25	1	HClO ₄	lgK:3.62 (3SF)	4	MAX	21702
4	25	1	NaClO ₄	lgK:3.6 (2SF)	3	MDS	437
5	25	1	NaClO ₄	lgK:3.75 (3SF)	3	MQH	437
6	25	1	NaClO ₄	lgK:3.75 (3SF)	3	MQH	931
7	25	1	NaClO ₄	lgK:3.75 (3SF)	3	MQH	30109
8	25	3	HClO ₄	lgK:4.90 (3SF)	3	MDS	437

Reaction No. 54410



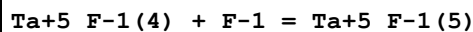
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO ₄	lgK:3.3 (2SF)	4	MAX	140
2	25	1	HClO ₄	lgK:3.34 (3SF)	4	MAX	21702
3	25	1	NaClO ₄	lgK:3.3 (2SF)	3	MAX	437
4	25	1	NaClO ₄	lgK:3.10 (3SF)	3	MQH	437
5	25	1	NaClO ₄	lgK:3.10 (3SF)	3	MQH	931
6	25	1	NaClO ₄	lgK:3.10 (3SF)	3	MQH	30109
7	25	3	NaClO ₄	lgK:4.48 (3SF)	3	MDS	437

Reaction No. 54411



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO ₄	lgK:3.0 (2SF)	4	MAX	140
2	25	1	HClO ₄	lgK:2.97 (3SF)	4	MAX	21702
3	25	1	NaClO ₄	lgK:0.66 (2SF)	0	MQH	931
4	25	1	NaClO ₄	lgK:0.66 (2SF)	0	MQH	30109

Reaction No. 54484



Note: Reacting species unknown, probably hydroxylated

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO ₄	lgK:4.8 (2SF)	2	MAX	140
2	25	1	HClO ₄	lgK:4.81 (3SF)	2	MAX	21702
3	25	1	NaClO ₄	lgK:4.8 (2SF)	2	MAX	437
4	25	3	HClO ₄	lgK:4.91 (3SF)	2	MDS	437

Reaction No. 58550

Ta+5_F-1(8) + F-1 = Ta+5_F-1(9)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO ₄	lgK:3.6(2SF)	3	MAX	140
2	25	1	HClO ₄	lgK:3.57(3SF)	4	MAX	21702

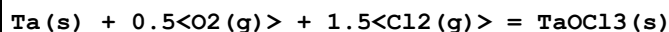
Reaction No. 66379							
Cl ₂ (g) + 0.5<O ₂ (g)> + Ta(s) = TaOCl ₂ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-191.(3SF)	2	CRV	22971
2	25	0	Inf. Dilution	dH:-213.(3SF)	2	CRV	22971

Reaction No. 66380							
Ta(s) + 2<S(s)> = Ta+4_S-2(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:60.43(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:60.05(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:44.62(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:35.23(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:28.92(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-82.44(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-84.6(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-85.7(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-86.3(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-86.8(3SF)	0	MTD	6422

Reaction No. 66381							
Ta(s) + O ₂ (g) + 0.5<Cl ₂ (g)> = TaO ₂ Cl(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:169.7(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:168.5(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:123.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:95.82(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:77.71(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-231.5(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-250.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-249.4(4SF)	3	MTD	6422
9	227	0	Inf. Dilution	dH:-248.8(4SF)	2	MTD	6422

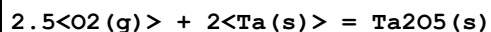
10	327	0	Inf. Dilution	dH:-248.2(4SF)	0	MTD	6422
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Reaction No. 66382



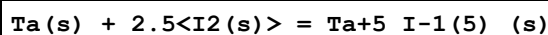
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:140.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:139.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:100.9(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:77.67(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:62.27(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-191.8(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-213.3(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-212.6(4SF)	2	MTD	6422
9	227	0	Inf. Dilution	dH:-211.8(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-211.0(4SF)	0	MTD	6422

Reaction No. 66383



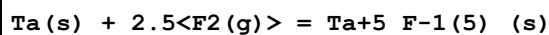
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:334.8(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:332.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:243.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:190.2(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:154.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-456.7(4SF)	1	EFE	6422
7	25	0	Inf. Dilution	dG:-457.0(4SF)	4	CRV	22971
8	25	0	Inf. Dilution	dG:-460.2(4SF)	2	CRV	28287
9	25	0	Inf. Dilution	dG:-456.5(0.6SD)	4	EFE	30139
10	25	0	Inf. Dilution	dH:-489.0(4SF)	2	MTD	6422
11	25	0	Inf. Dilution	dH:-489.2(4SF)	4	CRV	22971
12	25	0	Inf. Dilution	dH:-488.8(0.5SD)	4	MCL	30139
13	127	0	Inf. Dilution	dH:-488.6(4SF)	1	MTD	6422
14	227	0	Inf. Dilution	dH:-488.1(4SF)	0	MTD	6422
15	327	0	Inf. Dilution	dH:-487.4(4SF)	0	MTD	6422

Reaction No. 66384



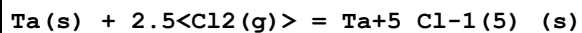
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.90 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:51.58 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:38.64 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:28.86 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:21.25 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-70.8 (3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-70.0 (3SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-79.7 (3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-105.1 (4SF)	1	MTD	6422
10	327	0	Inf. Dilution	dH:-103.7 (4SF)	0	MTD	6422

Reaction No. 66385



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:313.8 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:311.7 (4SF)	0	ENS	6422
3	95	0	Inf. Dilution	lgK:250.5 (4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-428.0 (4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dG:-428. (3SF)	4	CRV	22971
6	25	0	Inf. Dilution	dH:-454.9 (4SF)	4	MTD	6422
7	25	0	Inf. Dilution	dH:-455.0 (4SF)	4	CRV	22971
8	95	0	Inf. Dilution	dH:-454.0 (4SF)	4	MTD	6422

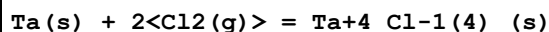
Reaction No. 66386



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:130.8 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:129.8 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:92.52 (4SF)	2	ENS	6422
4	216.7	0	Inf. Dilution	lgK:72.10 (4SF)	1	ENS	6422
5	25	0	Inf. Dilution	dG:-178.4 (4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dG:-179. (3SF)	4	CRV	22971
7	25	0	Inf. Dilution	dH:-205.3 (4SF)	2	MTD	6422
8	25	0	Inf. Dilution	dH:-205.5 (4SF)	4	CRV	22971
9	127	0	Inf. Dilution	dH:-204.4 (4SF)	1	MTD	6422

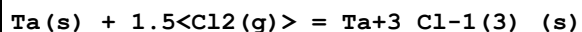
10	216.7	0	Inf. Dilution	dH:-203.7(4SF)	0	MTD	6422
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Reaction No. 66387



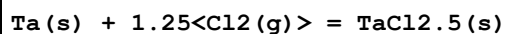
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:107.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:106.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:76.27(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:58.05(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:45.96(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-146.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-167.7(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-167.0(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-166.3(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-165.6(4SF)	0	MTD	6422

Reaction No. 66388



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:84.10(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:83.51(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:59.77(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:45.59(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:36.17(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-114.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-130.5(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-130.1(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-129.6(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-129.1(4SF)	0	MTD	6422

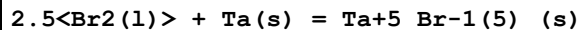
Reaction No. 66389



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:74.54(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:74.03(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:53.20(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:40.75(4SF)	1	ENS	6422

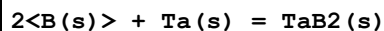
5	327	0	Inf. Dilution	lgK:32.47 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-101.7 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-114.5 (4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-114.1 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-113.8 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-113.4 (4SF)	0	MTD	6422

Reaction No. 66390



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:95.84 (4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:95.20 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:67.17 (4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:49.71 (4SF)	0	ENS	6422
5	240	0	Inf. Dilution	lgK:47.95 (4SF)	2	ENS	6422
6	25	0	Inf. Dilution	dG:-130.8 (4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dG:-132. (3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-598.3 (4SF) kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-143.0 (4SF)	3	MTD	6422
10	25	0	Inf. Dilution	dH:-143.0 (4SF)	4	CRV	22971
11	127	0	Inf. Dilution	dH:-160.4 (4SF)	0	MTD	6422
12	227	0	Inf. Dilution	dH:-159.0 (4SF)	0	MTD	6422
13	240	0	Inf. Dilution	dH:-158.8 (4SF)	0	MTD	6422

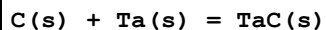
Reaction No. 67233



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.19 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:35.97 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:26.86 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:21.40 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:17.75 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-49.37 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-50.0 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-50.0 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-50.0 (3SF)	4	MTD	6422

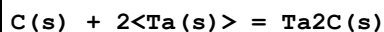
10	327	0	Inf. Dilution	dH:-50.0 (3SF)	0	MTD	6422
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Reaction No. 67234



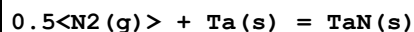
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.99 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:24.84 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:18.57 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:14.82 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:12.33 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-34.09 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-35.0 (3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-34.44 (4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-35.5 (3SF)	3	CRV	22971
10	25	0	Inf. Dilution	dH:-38.5 (0.6SD)	3	MCL	30139
11	127	0	Inf. Dilution	dH:-34.36 (4SF)	3	MTD	6422
12	227	0	Inf. Dilution	dH:-34.27 (4SF)	3	MTD	6422
13	327	0	Inf. Dilution	dH:-34.19 (4SF)	0	MTD	6422

Reaction No. 67235



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.12 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:36.89 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:27.61 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:22.04 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:18.34 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-50.64 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-51.0 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-50.9 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-50.8 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-50.8 (3SF)	0	MTD	6422

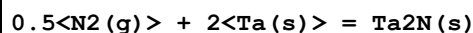
Reaction No. 67236



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.69 (4SF)	4	ENS	6422

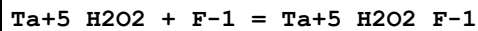
2	27	0	Inf. Dilution	lgK:39.42 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:28.45 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:21.88 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:17.53 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-54.15 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-53. (2SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-60.30 (4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-60. (2SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-60.17 (4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-59.95 (4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-59.68 (4SF)	0	MTD	6422

Reaction No. 67237



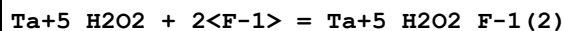
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.73 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:42.44 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:30.63 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:23.56 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:18.86 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-58.30 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-64.90 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-64.79 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-64.60 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-64.37 (4SF)	0	MTD	6422

Reaction No. 68864



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4 (1) :Water 20:80Vol	lgK:2.39 (3SF)	0	MSP	6680

Reaction No. 68865



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4 (1) :Water 20:80Vol	lgK:2.35 (3SF)	0	MSP	6680

Reaction No. 68866							
$\text{Ta+5_H2O2} + 3\text{<F-1>} = \text{Ta+5_H2O2_F-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4(1):Water 20:80Vol	lgK:1.93(3SF)	0	MSP	6680

Reaction No. 68867							
$\text{Ta+5_H2O2} + 4\text{<F-1>} = \text{Ta+5_H2O2_F-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4(1):Water 20:80Vol	lgK:1.83(3SF)	0	MSP	6680

Reaction No. 70988							
$\text{Ta+3} + 3\text{<e-1>} = \text{Ta(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.6(1SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:-0.6(1SF)	4	CRV	22519

Reaction No. 70989							
$\text{Ta2O5(s)} + 10\text{<H+1>} + 10\text{<e-1>} = 2\text{<Ta(s)>} + 5\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.81(2SF)	0	MDG	140
2	25	0	Inf. Dilution	E:-0.750(3SF)	2	CRV	6330
3	25	0	Inf. Dilution	E:-0.750(3SF)	2	CRV	22519
4	25	0.1	Unknown	E:-0.81(2SF)	0	CRV	22551
Note (4): Phase of Ta2O5 not specified in source							

Reaction No. 73376							
$\text{Ta2O5(s)} + 5\text{<H2O>} = 2\text{<Ta+5_OH-1(5)>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-20.(2SF)	1	EES	
Note (1): Estimate based on Peiffer (Nb) and Timofeev (Nb & Ta for high T)							
2	100	3	NaClO4	lgK:-17.4(3SF)	4	MSL	30118
3	150	3	NaClO4	lgK:-17.1(3SF)	4	MSL	30118
4	200	3	NaClO4	lgK:-16.4(3SF)	4	MSL	30118
5	250	3	NaClO4	lgK:-16.4(3SF)	3	MSL	30118

Reaction No. 74263							
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Ta+5_F-1(7) + 5<e-1> = Ta(s) + 7<F-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-84.44(4SF)	1	ELC	
2	25	0	Unknown	E:-0.45(2SF)	0	CRV	22551
Note (2): Concentration / ionic strength not specified in source							

Reaction No. 77489							
O2(g) + Ta(s) - e-1 = TaO2+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-201.4(4SF)	0	CRV	28287
Note (1): Species in aqueous solution will be hydrated							

Reaction No. 77490							
0.5<O2(g)> + Ta(s) - 2<e-1> = TaO+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-121.65(5SF)	3	CRV	28287

Reaction No. 77494							
Ta+2 + 2<e-1> = Ta(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.225(3SF)	3	CRV	28287

Reaction No. 77495							
Ta+3 + e-1 = Ta+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.35(3SF)	3	CRV	28287

Reaction No. 77496							
TaO2(s) + e-1 + 4<H+1> = Ta+3 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.8(1SF)	0	CRV	28287
Note (1): Probably, it's Ta2O5(fresh,s) and not Ta2O5(s)							

Reaction No. 79586							
Ta+5_OH-1(5)_(s) = Ta+5_OH-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:-5.2(2SF)	3	EOD	403

Note (1): Calc. from Babko solubility [63BLN_30106]							
2	25	1	KNO3	lgK:-5.355(4SF)	3	EOD	7093
Note (2): Calc. from Babko solubility [63BLN_30106]; $\text{Ta}_2\text{O}_5(\text{s}) + 5\text{H}_2\text{O} = 2\text{Ta}_5\text{OH-1(5)}$ with lgK=-10.71, here divided by 2							

Reaction No. 80122							
$\text{Ta}_6\text{O}_{19-8} + \text{H}^+ = \text{H}^+_{\text{Ta}_6\text{O}_{19-8}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.02(0.03SD)	2	MGL	7093
Note (1): Application of Bromley theory							
2	25	3	KCl	lgK:13.89(0.02SD)	0	MGL	7093
Note (2): Above MGL limits; pH from 12.4 to 13							
3	25	3	KCl	lgK:11.5(3SF)	2	MMR	30236
Note (3): pH from 10.4 to 14.5							

Reaction No. 80123							
$\text{Ta}_6\text{O}_{19-8} + 2\text{H}^+ = \text{H}^+_{(2)}\text{Ta}_6\text{O}_{19-8}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.35(0.03SD)	3	MGL	7093
Note (1): Application of Bromley theory							
2	25	3	KCl	lgK:25.91(0.01SD)	1	MGL	7093
Note (2): pH from 12.4 to 13; first pK above MGL limits							

Reaction No. 80124							
$\text{Ta}_5\text{OH-1(5)} = \text{Ta}_5\text{OH-1(4)} + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:-13.0(3SF)	3	MSL	30106
2	25	1	KNO3	lgK:-13.0(3SF)	1	EES	

Reaction No. 80125							
$\text{Ta}_5\text{OH-1(5)} + \text{H}_2\text{O} = \text{Ta}_5\text{OH-1(6)} + \text{H}^+$							
Note: Basic pK for $\text{Ta}(\text{OH})_5$ might not really exist if polyoxotantalates formed							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:-9.6(2SF)	0	MSL	403
2	19	1	KNO3	lgK:-9.6(2SF)	0	MSL	30106
3	25	1	KNO3	lgK:-9.6(2SF)	1	EES	

Reaction No. 80126							
$\text{Ta+5_OH-1(5)} + \text{H+1} = \text{Ta+5_OH-1(4)} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:1.(1SF)	3	MSL	403
2	25	1	KNO3	lgK:1.(1SF)	1	EES	

Reaction No. 80127							
$\text{Ta6O19-8} + \text{H2O} = \text{H+1_Ta6O19-8} + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:12.68(0.1SD)	3	MGL	30159
Note (1): pH from 8.5 to 12							

Reaction No. 80128							
$\text{H+1(2)_Ta6O19-8} + \text{H2O} = \text{H+1(3)_Ta6O19-8} + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:9.28(0.05SD)	3	MGL	30159
Note (1): pH from 8.5 to 12							

Reaction No. 80129							
$\text{Ta2O5(s)} + 6\text{<H+1_F-1>} + \text{H2O} = 2\text{<Ta+5_F-1(3)_OH-1(3)>} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.(2SF)	1	EDH	
Note (1): Guessing desparately							
2	100	3	NaClO4	lgK:-8.24(3SF)	5	MSL	30118
3	150	3	NaClO4	lgK:-7.45(3SF)	5	MSL	30118
4	200	3	NaClO4	lgK:-8.60(3SF)	5	MSL	30118
5	250	3	NaClO4	lgK:-8.55(3SF)	5	MSL	30118

Reaction No. 80130							
$\text{Ta+5_F-1(3)} + 3\text{<F-1>} = \text{Ta+5_F-1(6)}$							
Note: Reacting species unknown, probably hydroxylated							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.8(3SF)	1	ELC	
2	25	3	HClO4	lgK:15.7(3SF)	2	MDS	30162

Reaction No. 80131							
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Ta+5_F-1(3) + 4<F-1> = Ta+5_F-1(7)							
Note: Reacting species unknown, probably hydroxylated							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.7 (3SF)	1	ELC	
2	25	3	HClO4	lgK:20.15 (4SF)	2	MDS	30162

Reaction No. 80133							
H+1_Ta6O19-8 + H2O = H+1(2)_Ta6O19-8 + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:10.81 (0.1SD)	3	MGL	30159
Note (1): pH from 8.5 to 12							

Reaction No. 80134							
H+1_Ta6O19-8 + H+1 = H+1(2)_Ta6O19-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14. (2SF)	1	EES	
Note (1): Guess based on Ta6O19-8 + 2<H+1> = H+1(2)_Ta6O19-8 (ex Bromley)							
2	25	3	KCl	lgK:9.3 (2SF)	4	MMR	30236
Note (2): pH from 10.4 to 14.5							

Reaction No. 80135							
3<Cl2(g)> + Cs(s) + Ta(s) = Ta+5_Cs+1_Cl-1(6)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-337. (3SF)	4	CRV	22971

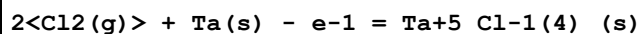
Reaction No. 80136							
3<Cl2(g)> + Rb(s) + Ta(s) = Ta+5_Rb+1_Cl-1(6)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-326. (3SF)	4	CRV	22971

Reaction No. 80137							
3<Cl2(g)> + K(s) + Ta(s) = Ta+5_K+1_Cl-1(6)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-320. (3SF)	4	CRV	22971

Reaction No. 80138							
3<Cl2(g)> + Na(s) + Ta(s) = Ta+5_Na+1_Cl-1(6)_(s)							

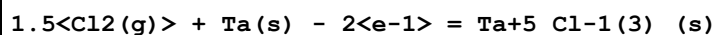
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-308.(3SF)	4	CRV	22971

Reaction No. 80139



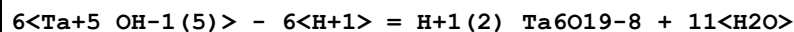
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-149.(3SF)	4	CRV	22971
2	25	0	Inf. Dilution	dH:-170.(3SF)	4	CRV	22971

Reaction No. 80140



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-123.(3SF)	4	CRV	22971
2	25	0	Inf. Dilution	dH:-139.(3SF)	4	CRV	22971

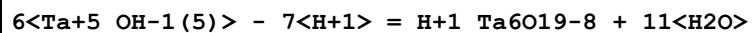
Reaction No. 80159



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KNO3	lgK:-32.12(4SF)	3	EOD	7093

Note (1): Recalculated own data + solubility Babko (30106)

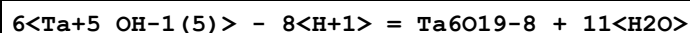
Reaction No. 80160



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KNO3	lgK:-44.02(4SF)	3	EOD	7093

Note (1): Recalculated own data + solubility Babko (30106)

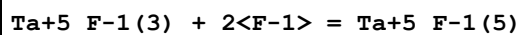
Reaction No. 80161



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KNO3	lgK:-57.48(4SF)	3	EOD	7093

Note (1): Recalculated own data + solubility Babko (30106)

Reaction No. 80169

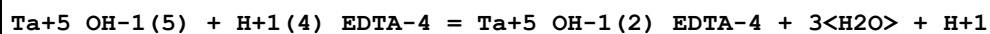


Note: Reacting species unknown, probably hydroxylated

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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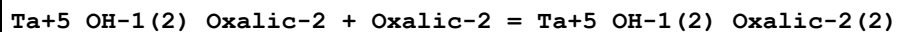
1	25	1	NaClO ₄	lgK:10.80 (4SF)	2	MDS	5398
2	25	3	HClO ₄	lgK:10.77 (4SF)	2	MDS	30162

Reaction No. 80181



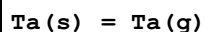
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1.08	K ₂ SO ₄	lgK:33.6 (3SF)	3	MPL	30359
2	25	0	Inf. Dilution	lgK:34. (2SF)	1	EES	

Reaction No. 80196



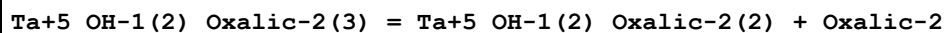
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0	Self Medium	lgK:7.4 (2SF)	3	MSL	30107
2	25	0	Inf. Dilution	lgK:7.5 (2SF)	1	EES	

Reaction No. 80206



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:176.67 (5SF)	3	CRV	31291
2	25	0	Inf. Dilution	dH:186.90 (5SF)	3	MTD	31291

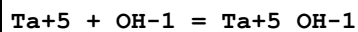
Reaction No. 80209



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Self Medium	lgK:5.91 (0.05SD)	2	MSL	30164

Note (1): pH 0.3-1.8

Reaction No. 80211



Note: Ta+5 is considered too highly charged to exist

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO ₃	lgK:29.6 (3SF)	0	EQC	30106

Note (1): Highly dubious method

2	25	1	KNO ₃	lgK:29.6 (3SF)	0	EES	
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Reaction No. 80212

Ta+5_OH-1 + OH-1 = Ta+5_OH-1(2)							
Note: Ta+5_OH-1 is considered too highly charged to exist							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:26.6(3SF)	0	EQC	30106
Note (1): Highly dubious method							
2	25	1	KNO3	lgK:26.6(3SF)	0	EES	

Reaction No. 80213							
Ta+5_OH-1(2) + OH-1 = Ta+5_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:22.4(3SF)	0	EQC	30106
Note (1): Highly dubious method							
2	25	1	KNO3	lgK:22.4(3SF)	1	EES	

Reaction No. 80214							
Ta+5_OH-1(3) + OH-1 = Ta+5_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	KNO3	lgK:17.7(3SF)	0	EQC	30106
Note (1): Highly dubious method							
2	25	1	KNO3	lgK:17.7(3SF)	1	EES	

Reaction No. 80525							
Ta2O5(s) + 10<H+1_F-1> = 2<Ta+5_F-1(5)> + 5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.13(2SF)	1	EES	
2	100	3	NaClO4	lgK:0.13(2SF)	5	MSL	30118
3	150	3	NaClO4	lgK:0.35(2SF)	5	MSL	30118

Reaction No. 80596							
1.5<O2(g)> + 2<Ta(s)> = Ta2O3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-459.71(5SF)	3	CRV	31291

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EQC	Quintuply charged central ion method
ESC	Standard conditions anchor
MAX	Anion exchange
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MGL	Glass electrode
MIS	Ion selective electrode
MMR	Nuclear magnetic resonance
MPL	Polarography
MQH	Quinhydrone electrode
MSL	Solubility

MSP	Spectroscopy
MTD	Temperature difference

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